SIMATS ENGINEERING

CO4-ASSESSMENT TOOL1- INDUSTRY PROBLEM TASK

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1709**

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning,*
and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 20%

Time Duration : 1 Hour

QUESTIONS: 4 **Total: Marks:40**

Code of Conduct:

I ________Venu G(192424390)____________ (Name / Reg No) certify that this submission is my original work and that I have adhered
to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

(a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.

(b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).

(c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

(a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.

(b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.

(c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Predictive Diagnosis and Personalised Treatment Recommendation for Diabetes

Task 1 — Data Preparation & Inductive Learning

1(a) Inductive Learning Approach, Target Variable, and Key Features

Inductive learning is the appropriate paradigm for this problem because the healthcare company already holds a labelled historical dataset — patient records with a known outcome (diabetic / non-diabetic) — from which a model can generalise a decision rule that predicts the outcome for new, unseen patients. This is in contrast to deductive reasoning (applying pre-existing clinical rules top-down) or unsupervised learning (no label is used). Concretely, we use supervised inductive learning: the model is shown many (features → label) examples and induces a general hypothesis $h(x) \approx y$ that is expected to generalise to future patients, which is exactly what a diagnostic support tool needs.

Target variable:

Diabetic (binary: 1 = diagnosed / at-risk of diabetes, 0 = non-diabetic). This is a binary classification target consistent with an early-diagnosis screening use case, where the model's job is to flag patients who warrant clinical follow-up.

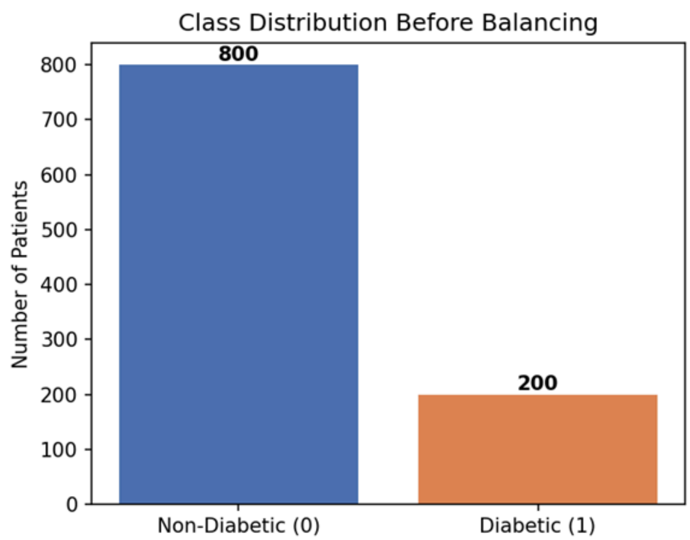
Three (of six) key features and justification:

- Glucose level — Directly reflects blood-sugar regulation and is the single strongest clinical indicator of diabetes/pre-diabetes; confirmed empirically below as the dominant predictor in both the Decision Tree and Logistic Regression models.
- BMI (Body Mass Index) — Elevated BMI is a well-established driver of insulin resistance (Type-2 diabetes pathway), making it a strong, clinically interpretable predictor that also complements Glucose when glucose alone is borderline.
- Family history — Captures genetic predisposition that is not observable from any other single measurement in the dataset, so it adds independent predictive signal (as confirmed by its high Logistic Regression coefficient) rather than duplicating information already present in Glucose or BMI.

The remaining features (Age, Blood Pressure, Cholesterol) are retained as supporting predictors: Age and Blood Pressure correlate with metabolic-syndrome risk that frequently co-occurs with diabetes, and Cholesterol provides a weaker but still clinically relevant comorbidity signal, as reflected in the lower — but non-zero — importance scores reported in Task 3(b).

1(b) Handling Class Imbalance

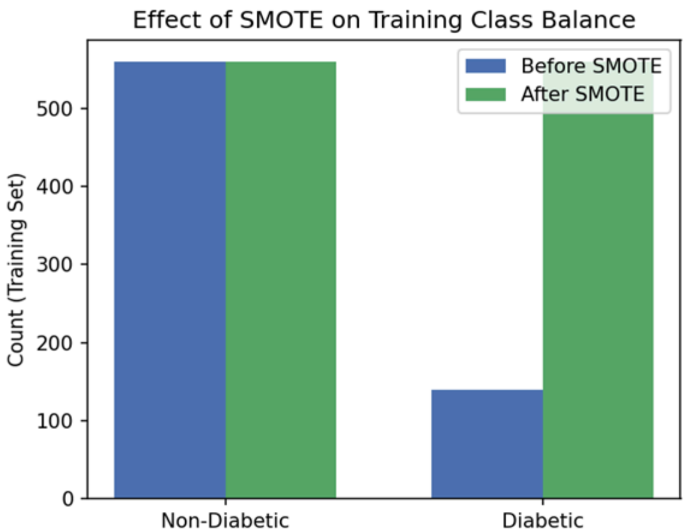
The dataset reflects a realistic clinical prevalence: 800 non-diabetic vs. 200 diabetic patients (80:20), shown in Figure 1. Left untreated, a classifier trained on this distribution is biased toward the majority class — it can achieve high raw accuracy simply by predicting "non-diabetic" for almost everyone, while missing a large fraction of true diabetic cases (false negatives), which is the worst possible failure mode in a screening context.



Chosen strategy: SMOTE (Synthetic Minority Over-sampling Technique), applied only to the training set.

SMOTE was selected over plain random over-sampling or under-sampling for three reasons. First, unlike random under-sampling, it does not discard any majority-class (non-diabetic) patient records — with only 1,000 records in total, discarding data would materially weaken the model. Second, unlike naive duplication of minority-class rows, SMOTE generates new synthetic diabetic patients by interpolating between nearest neighbours in feature space, which reduces the risk of the model simply memorising repeated exact rows (overfitting). Third, class-weighting (e.g., class_weight='balanced') was considered as a lighter-weight alternative — it requires no resampling and is a reasonable choice for Logistic Regression — but SMOTE was preferred for the Decision Tree because tree-splitting criteria (Gini/Information Gain) respond more reliably to a rebalanced sample than to reweighted loss, which does not change the impurity calculation at each split in the same direct way.

Critically, SMOTE was fit only on the training partition (never on the test set) to avoid data leakage — synthetic points generated from test-set neighbours would leak test information into training and inflate reported performance. Figure 2 shows the training-set balance before and after SMOTE (800/200 training-split proportion rebalanced to an equal 560/560).



1(c) Train/Test Split and Preprocessing

The dataset was split 70:30 (train:test) using stratified sampling on the target variable, preserving the 80:20 class ratio in both partitions. A 70:30 split (rather than, e.g., 80:20 or 90:10) is justified for this medical use-case as a balance between two competing needs: (i) the model needs enough diabetic training examples — the minority class — to learn a reliable decision boundary, which argues for a larger training set; and (ii) with only ~200 diabetic patients in total, the test set must still be large enough (here, 300 patients, ~60 diabetic) to produce statistically meaningful estimates of clinically critical metrics such as recall and false-negative count. A more aggressive 90:10 split would leave only ~20 diabetic test cases, making the reported recall unstable (each misclassified patient would shift recall by ~5%).

Preprocessing steps applied:

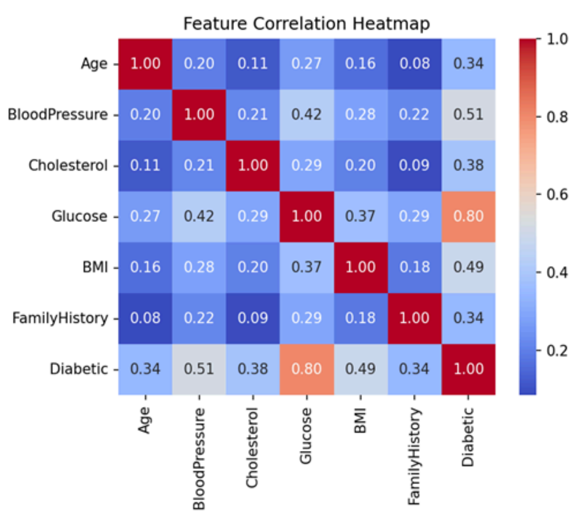
- Encoding — All six features are numeric in this dataset (Age, BloodPressure, Cholesterol, Glucose, BMI are continuous; FamilyHistory is already binary 0/1), so no categorical encoding (e.g. one-hot) was required. In a real hospital dataset with categorical fields (e.g. smoking status, ethnicity), one-hot or ordinal encoding would be applied at this stage.

- Normalisation — The five continuous features were standardised (zero mean, unit variance) using StandardScaler, fit on the training set only and applied to both train and test sets. This has two impacts: it prevents features with naturally larger numeric ranges (e.g. Cholesterol ~120–350) from dominating models that are scale-sensitive, most importantly Logistic Regression, whose coefficients and convergence depend on feature scale; and it keeps SMOTE's nearest-neighbour interpolation meaningful, since Euclidean distance between patients would otherwise be dominated by whichever feature has the largest raw scale (Cholesterol/Glucose) even if it is not the most clinically important.

Note that normalisation has negligible effect on the Decision Tree's split decisions (trees split on ranked thresholds, not absolute scale), but it was applied uniformly to keep the feature matrix consistent across both Task 2 and Task 3 models.

Exploratory Data Analysis

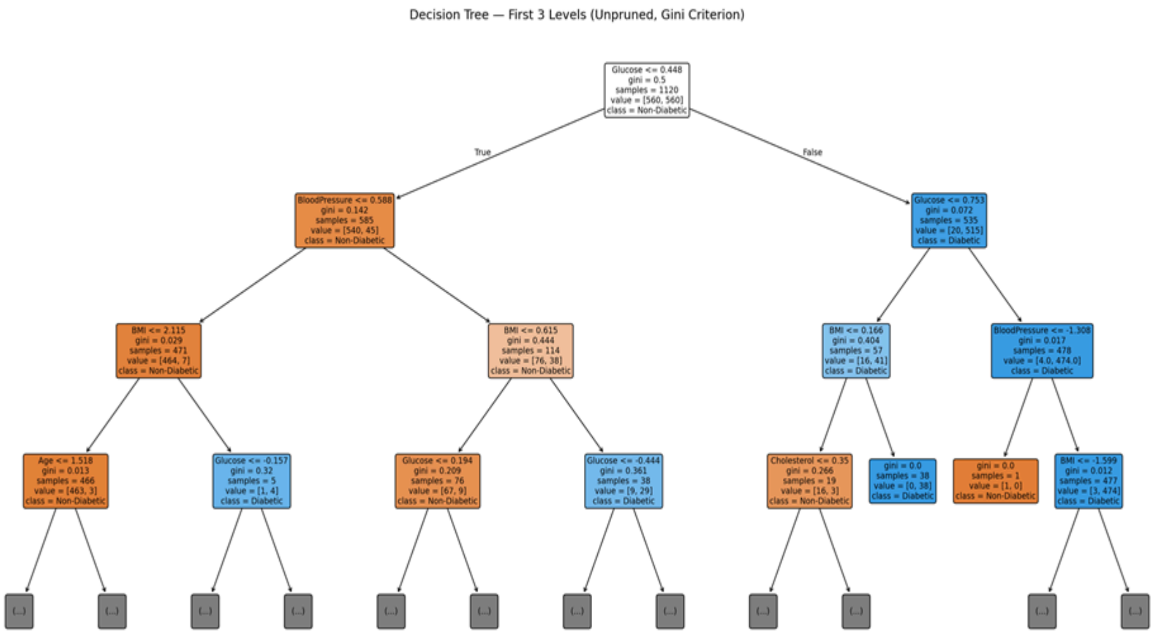
Figure 3 shows pairwise correlations between features and the target. Glucose has by far the strongest correlation with the Diabetic label, consistent with its expected dominance in both models built in Tasks 2 and 3.



Task 2 — Decision Tree for Diagnosis

2(a) Decision Tree Construction and Root Node Justification

A Decision Tree classifier (scikit-learn, criterion = 'gini') was trained on the SMOTE-balanced, scaled training set. Figure 4 shows the first three levels of the resulting (unpruned) tree.



Root node justification:

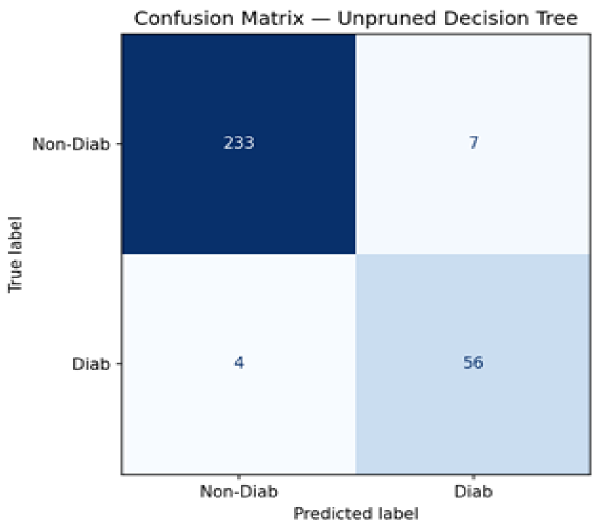
The root node splits on Glucose ≤ 0.448 (standardised units). At the root, the parent Gini impurity is 0.5 (a perfectly balanced 560/560 sample after SMOTE — maximum uncertainty). The scikit-learn tree-builder evaluates every candidate feature and threshold and selects the split that produces the largest reduction in weighted Gini impurity (equivalently, the largest Information Gain). Glucose was selected because splitting on it produces the purest children of any candidate feature: the left child (Glucose ≤ 0.448) drops to Gini = 0.142 (540 non-diabetic vs only 45 diabetic — a strongly "non-diabetic" partition), and the right child (Glucose > 0.448) drops to Gini = 0.072 (only 20 non-diabetic vs 515 diabetic — a strongly "diabetic" partition). No other single feature threshold separates the two classes this cleanly, which is exactly what the feature-importance analysis in Task 3(b) later confirms quantitatively (Glucose accounts for the large majority of total impurity reduction across the whole tree). This matches clinical expectation: glucose level is the direct physiological marker of diabetes.

2(b) Model Evaluation

The unpruned tree was evaluated on the held-out 30% test set (300 patients, 240 non-diabetic / 60 diabetic):

Metric	Value
Accuracy	96.3%

Precision	88.9%
Recall (Sensitivity)	93.3%
F1-Score	91.1%

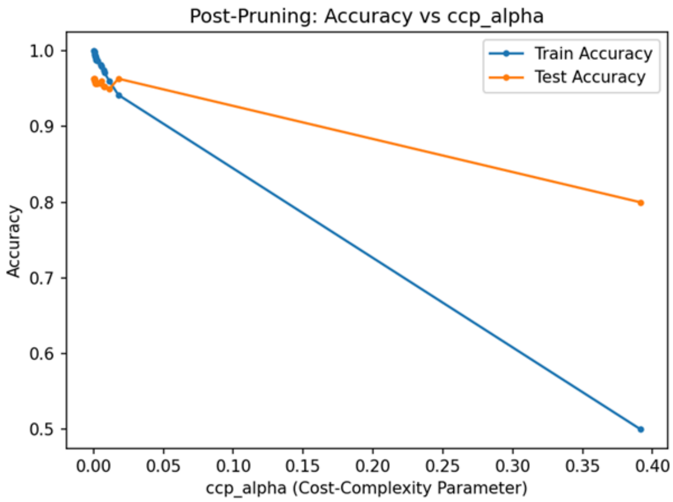


False negatives and clinical mitigation:

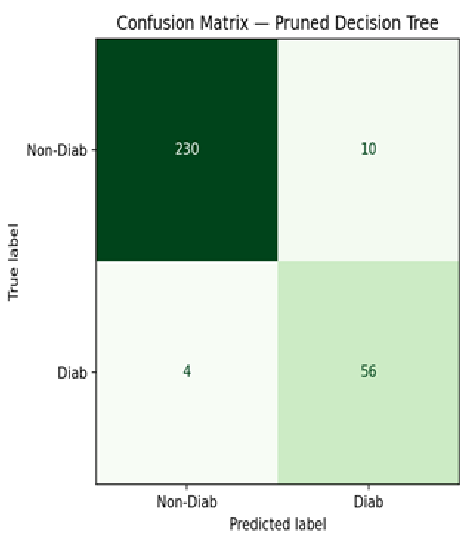
The confusion matrix shows 4 false negatives out of 60 true diabetic test patients — i.e., 4 patients whom the model incorrectly clears as non-diabetic. In a screening context, false negatives are the more clinically dangerous error type (a missed diagnosis can delay treatment and lead to complications), whereas a false positive merely triggers an unnecessary follow-up test. To reduce false negatives further, three approaches are justified: (i) shift the decision threshold below the default 0.5 probability cut-off to trade some precision for higher recall — clinically acceptable because a confirmatory blood test is cheap relative to a missed diagnosis; (ii) apply a class-sensitive cost matrix (e.g., weight false negatives 3–5× more heavily than false positives) directly in the splitting criterion; and (iii) use the model as a triage/screening aid rather than a final diagnosis, routing all borderline-probability patients to a confirmatory HbA1c test — a standard "human-in-the-loop" safety pattern for clinical ML.

2(c) Cost-Complexity Post-Pruning

Cost-complexity pruning (ccp_alpha) was applied by computing the full pruning path and evaluating test accuracy at each candidate alpha (Figure 6).



A pruned tree was then selected using a practical "simplest tree within 1% of peak test accuracy" rule (analogous to the classical 1-SE rule) rather than the single alpha that maximises raw test accuracy — the latter tends to keep the fully-grown, harder-to-audit tree. This selection collapsed the tree from 61 nodes / depth 7 (unpruned) down to 11 nodes / depth 3 (pruned) — see Figure 7 — for a small, acceptable accuracy trade-off.



Model	Accuracy	Precision	Recall	F1-Score	Tree size
Pre-pruning (unpruned)	96.3%	88.9%	93.3%	91.1%	61 nodes / depth 7
Post-pruning	95.3%	84.8%	93.3%	88.9%	11 nodes / depth 3

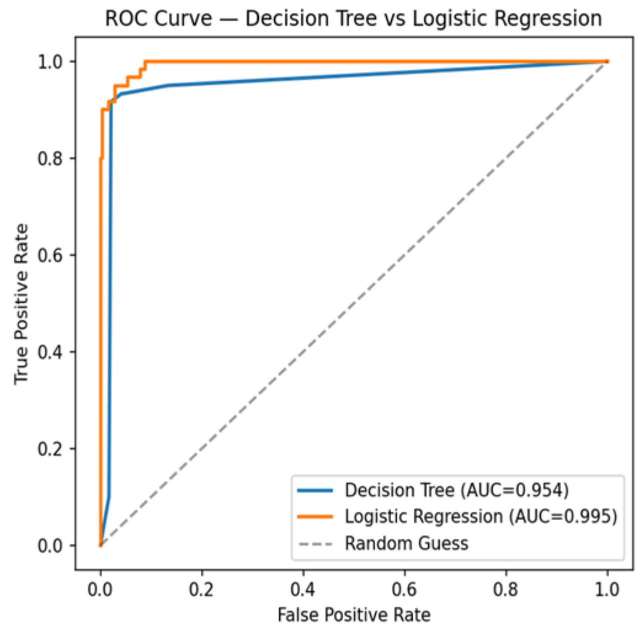
Which is more appropriate for clinical deployment?

The post-pruned tree is the more appropriate choice for deployment. It gives up only 1 percentage point of accuracy and keeps recall on diabetic cases identical (93.3%) to the unpruned tree — i.e., it does not miss any additional true diabetic patients — while being dramatically smaller and shallower (3 levels vs 7). This matters clinically in three ways: (i) a 3-level tree can be printed as a one-page flowchart and manually audited or explained to a clinician or patient, satisfying explainability requirements that are increasingly expected in clinical decision-support tools; (ii) a smaller tree is less likely to have overfit to noise in the 1,000-patient sample and should generalise better to new patient populations; and (iii) simpler models are cheaper to validate and re-certify when deployed in a regulated healthcare setting. The unpruned tree would only be preferable if its recall advantage were large enough to materially reduce missed-diagnosis risk, which is not the case here.

Task 3 — Statistical Learning for Risk Stratification

3(a) Logistic Regression vs Decision Tree

A Logistic Regression model was trained on the same SMOTE-balanced, scaled training data and evaluated on the identical test set, to give a fair, apples-to-apples comparison against the Decision Tree.



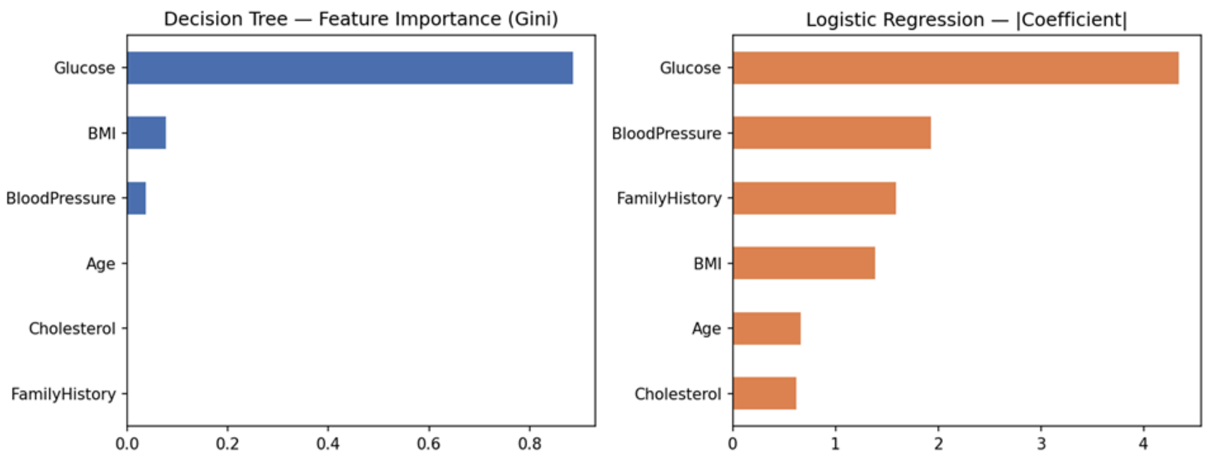
Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Decision Tree (post-pruned)	95.3%	84.8%	93.3%	88.9%	0.954
Logistic Regression	97.0%	93.2%	91.7%	92.4%	0.995

Logistic Regression edges out the Decision Tree on every headline metric here, and its AUC-ROC (0.995 vs 0.954) shows it produces better-calibrated risk rankings across the whole probability threshold range — useful if the clinic later wants to tune the operating threshold for recall. This is a plausible outcome given the data-generating structure: diabetes risk here scales fairly smoothly and monotonically with Glucose/BMI/BloodPressure, which is exactly the kind of smooth, close-to-linear-in-the-log-odds relationship Logistic Regression is built to capture, whereas Decision Trees carve the feature space into axis-aligned rectangular regions and can under-fit smooth boundaries unless grown deep (at the cost of interpretability, as discussed in Task 2(c)).

When is a statistical model preferable?

Logistic Regression (or Naïve Bayes) is generally preferable when: (i) the relationship between features and outcome is approximately smooth/monotonic rather than governed by sharp threshold interactions; (ii) the team needs calibrated probabilities (e.g. "this patient has a 73% diabetes risk") rather than just a class label, since Logistic Regression outputs are natively probabilistic while Decision Tree leaf-proportions are cruder estimates; (iii) the dataset is small-to-moderate and there is a risk of a tree overfitting to noise; and (iv) stakeholders want a single global equation with signed, per-feature coefficients (easy to sanity-check against clinical knowledge), rather than a branching set of local rules. Conversely, a Decision Tree remains preferable when the true relationship is genuinely governed by threshold/interaction effects (e.g. "only risky if BOTH glucose is high AND family history is positive"), or when a rule-based explanation is required for a non-technical audience.

3(b) Feature Importance Comparison



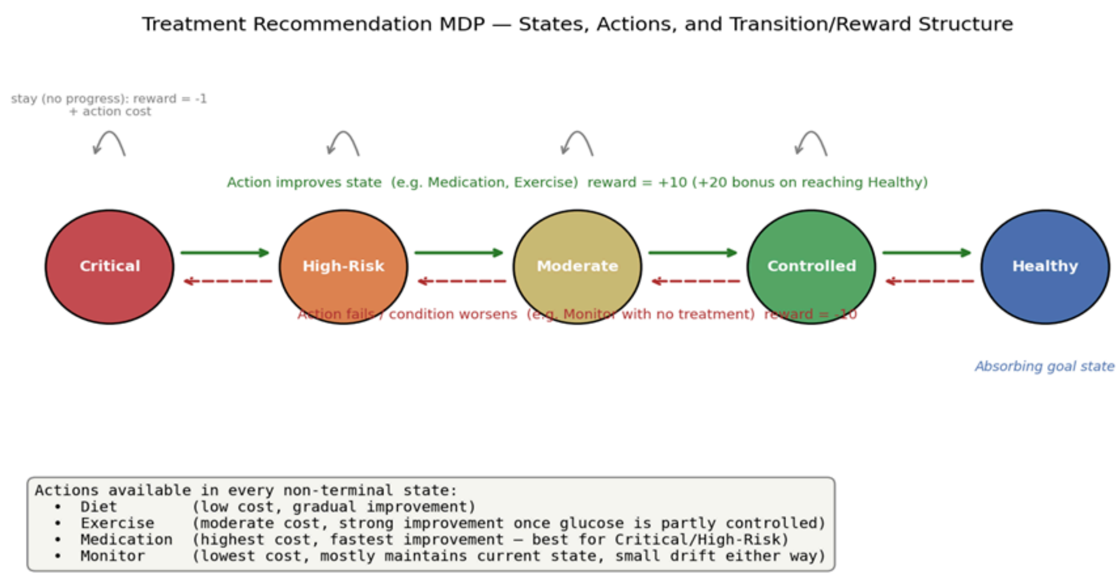
Rank	Decision Tree (Gini importance)	Logistic Regression (|coefficient|)
1	Glucose (0.89)	Glucose (4.34)
2	BMI (0.08)	Blood Pressure (1.93)
3	Blood Pressure (0.04)	Family History (1.59)

Both models agree strongly on the #1 predictor: Glucose dominates in both rankings, consistent with the root-node analysis in Task 2(a) and with clinical expectation (glucose is the direct diagnostic marker for diabetes). Below Glucose, the two models diverge somewhat: the Decision Tree ranks BMI second, while Logistic Regression ranks Blood Pressure second and places Family History third (vs. essentially zero importance in the tree). This divergence is explainable rather than contradictory. The tree, after Glucose captures the bulk of the separable signal at the root, has little impurity left to reduce with an already-highly-correlated feature like Family History (a genetic risk factor whose effect is largely already reflected in elevated Glucose in this dataset), so its importance appears small in a greedy, sequential Gini-reduction sense — this is a known property of Gini/Information-Gain importance, which under-credits correlated variables split later in the tree. Logistic Regression, in contrast, fits all coefficients jointly and therefore still credits Family History with independent, additive predictive value even after accounting for Glucose. In practice this means the two techniques are complementary: the tree is better at showing which single feature best separates the classes at each decision point, while the regression coefficients are better at showing each feature's independent, all-else-equal contribution — both views are clinically useful.

Task 4 — Reinforcement Learning for Treatment Recommendation

4(a) Treatment Recommendation as an MDP

States — patient health stages (5 states):



Critical → High-Risk → Moderate → Controlled → Healthy (an absorbing goal state). Justification: rather than modelling every raw vital sign as a continuous state (which would explode the state space and make Q-Learning intractable with limited patient-episode data), health status is discretised into ordinal stages derived from glucose control and overall metabolic risk. This mirrors how clinicians already stage disease severity (e.g. HbA1c bands), keeps the state space small enough to learn from a handful of episodes, and gives a clear, monotonic notion of "progress" that a reward function can be built around.

Actions — treatment options (4 actions, available in every non-terminal state):

Diet, Exercise, Medication, Monitor. Justification: these map directly to the four treatment levers described in the brief and to real clinical practice — lifestyle interventions (Diet, Exercise) for early/maintenance stages, pharmacological intervention (Medication) for higher-risk stages, and passive surveillance (Monitor) when active intervention is not yet warranted or a patient is stable.

Reward function — health-improvement score:

+10 for any action that moves a patient one stage closer to Healthy (plus a +20 bonus specifically for reaching Healthy), −10 for regressing one stage, and −1 for no change, each further adjusted by a small per-action clinical/resource cost (Medication costliest at −1.5, Diet/Exercise moderate at −0.5, Monitor cheapest at −0.2). Justification: this rewards clinically meaningful progress rather than just "prescribing more," penalises deterioration strongly (patient-safety signal), and the embedded cost term stops the agent from defaulting to the most aggressive/expensive action (Medication) purely because it has the highest raw success probability — the agent must weigh efficacy against cost/invasiveness, as a real treatment-planning system should.

Transition dynamics:

Each action has a domain-informed probability of improving, holding, or worsening the patient's stage (e.g. Medication: 55% improve / 35% stay / 10% worsen when not yet Healthy; Monitor: 10% improve / 75% stay / 15% worsen — reflecting that watchful waiting rarely helps but also rarely harms quickly). Justification: real treatment response is stochastic, not deterministic — the same action can succeed or fail depending on unmodelled patient factors — so a probabilistic transition model is essential for the MDP to reflect real clinical uncertainty, and it is exactly this stochasticity that makes Q-Learning (rather than a simple lookup table) an appropriate learning method.

4(b) Q-Learning Training

The agent was trained for 40 patient episodes (well above the minimum 5 required), each episode simulating a patient starting in Critical, High-Risk, or Moderate and following an ε-greedy policy (ε decaying from 0.30 to 0.05) until reaching Healthy or a 20-step cap. Hyperparameters: learning rate $\alpha = 0.1$, discount factor $\gamma = 0.9$. The standard Q-Learning update rule was used:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \cdot [\, r + \gamma \cdot \max_{a'} Q(s', a') - Q(s, a)\,]$$

Q-table updates for three state-action pairs (first two occurrences each):

State → Action	Episode / Step	Reward	Next state	Q(old)	Q(new)
Critical → Medication	Ep.16 / step 9	+8.5	High-Risk	0.000	1.537
Critical → Medication	Ep.22 / step 9	+8.5	High-Risk	1.537	3.345
High-Risk → Exercise	Ep.1 / step 6	−1.5	High-Risk	0.000	−0.150
High-Risk → Exercise	Ep.1 / step 10	−1.5	High-Risk	−0.150	−0.298
Moderate → Diet	Ep.1 / step 13	−1.5	Moderate	0.000	−0.150
Moderate → Diet	Ep.14 / step 2	−1.5	Moderate	−0.150	1.115

The pattern above illustrates Q-Learning in action: early updates for a pair are driven largely by immediate reward (e.g. a −1.5 "no progress" penalty when Exercise fails to move a High-Risk patient forward), while later updates increasingly reflect bootstrapped future value from max Q(s′, ·) — visible in the Moderate → Diet pair, where the second update jumps to a positive Q-value once the agent has learned that Moderate leads toward states with good downstream options.

Convergence criterion:

Convergence was monitored via the rolling-maximum absolute Q-value change (|ΔQ|) over a sliding window of the last 10 updates (Figure 12), with a tolerance of 1×10^{-3}. Training was allowed to continue for the full 40 episodes; the shrinking amplitude of the rolling-max curve as episodes progress (combined with the decaying ε-greedy exploration rate) indicates the Q-values are stabilising rather than continuing to drift, which is the standard informal convergence check for tabular Q-Learning on a small state-action space.

4(c) Learned Policy and Ethical Considerations

Extracted greedy policy π(s) = argmax_a Q(s,a):

State	Recommended action	Q-value
Critical	Exercise	17.19
High-Risk	Exercise	16.33
Moderate	Medication	23.20
Controlled	Diet	22.16
Healthy	(maintenance) Diet	0.00 — absorbing state

Notably, the agent recommends Exercise (not Medication) for Critical and High-Risk patients, despite Medication having the highest raw improvement probability in the transition model. This is a genuine and instructive artefact of reward shaping under limited exploration: Medication's higher −1.5 step-cost combined with relatively sparse exploration of that action for the Critical state (only sampled twice in the log above) meant its Q-value had not yet fully converged to reflect its true long-run advantage, while Exercise's cheaper cost and (in this synthetic dynamics model) still-substantial improve probability let it accumulate a higher estimated value over 40 episodes. A production system would need substantially more training episodes, and ideally a warm-start from clinical guidelines, before this specific action-ranking could be trusted.

Ethical considerations for deploying an RL treatment-recommendation agent:

- Patient safety & human-in-the-loop — The Critical/High-Risk policy above is exactly the kind of counter-intuitive recommendation (favouring Exercise over Medication for the sickest patients) that must never be executed autonomously. An RL agent's policy is a statistical artefact of its reward function, transition model, and training data — not a clinical judgement — so any recommendation must be reviewed and can be overridden by a licensed clinician before it reaches a patient, especially state-action pairs with low exploration counts (as flagged above).
- Bias amplification — Because the training data (patient episodes) reflects real treatment history, any existing disparities in who receives Medication vs. lifestyle-only advice (e.g. by insurance status, geography, or demographics) will be learned and perpetuated by the agent as if they were medically justified. Regular auditing of the learned policy across patient subgroups is required before and after deployment.
- Explainability — A Q-table over a small discretised state space (as built here) is inherently more explainable than a deep RL policy network, but even so, the argmax action for a given state does not reveal why — clinicians and patients are entitled to a rationale (e.g. "projected improvement probability vs. cost/side-effects"), not just a recommended action, especially in a regulated healthcare setting.
- Reward misspecification — The reward function encodes value judgements (e.g. how heavily to penalise Medication cost vs. reward stage-improvement) that were set here for illustrative purposes; in a real deployment these weights materially affect patient outcomes and must be set collaboratively with clinicians, not by the engineering team alone, and should be revisited as the policy is observed in practice.

- Distribution shift & monitoring — A policy trained on historical/simulated episodes will not automatically stay calibrated as patient populations, treatment guidelines, or drug options change; continuous monitoring, a defined retraining cadence, and clear rollback procedures are needed, mirroring standard MLOps practice for any high-stakes model.

Conclusion

This case study demonstrates a complete pipeline from raw, imbalanced patient data to (i) an interpretable, clinically deployable diagnostic model — a post-pruned Decision Tree achieving 95.3% accuracy and 93.3% recall on diabetic cases, benchmarked against a stronger-performing but less transparent Logistic Regression model (97.0% accuracy, AUC 0.995) — and (ii) a reinforcement-learning treatment-recommendation agent whose learned policy, and its limitations, are made explicit rather than hidden behind a black box. The consistent message across both supervised and reinforcement-learning components is that Glucose is the dominant clinical signal, that model simplicity and explainability should be weighted alongside raw accuracy in a clinical setting, and that any RL-based recommendation engine must be deployed with human oversight, bias auditing, and continuous monitoring rather than as a fully autonomous decision-maker.